Basic Principles of Medicine 2

Nervous System and Behavioral Sciences

Chemical Senses DLA: Gustation

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Learning Objectives

A student should be able to:

SOM.MK.I.BPM2.1.NB.1.NSCI.0269 Describe the basic taste qualities, and explain the functional anatomy of the gustatory system, focusing on taste buds and receptor cells, and the different signal transduction processes

SOM.MK.I.BPM2.1.NB.1.NSCI.0270 Describe the gustatory pathways including the location of cortical gustatory areas

The Gustatory System

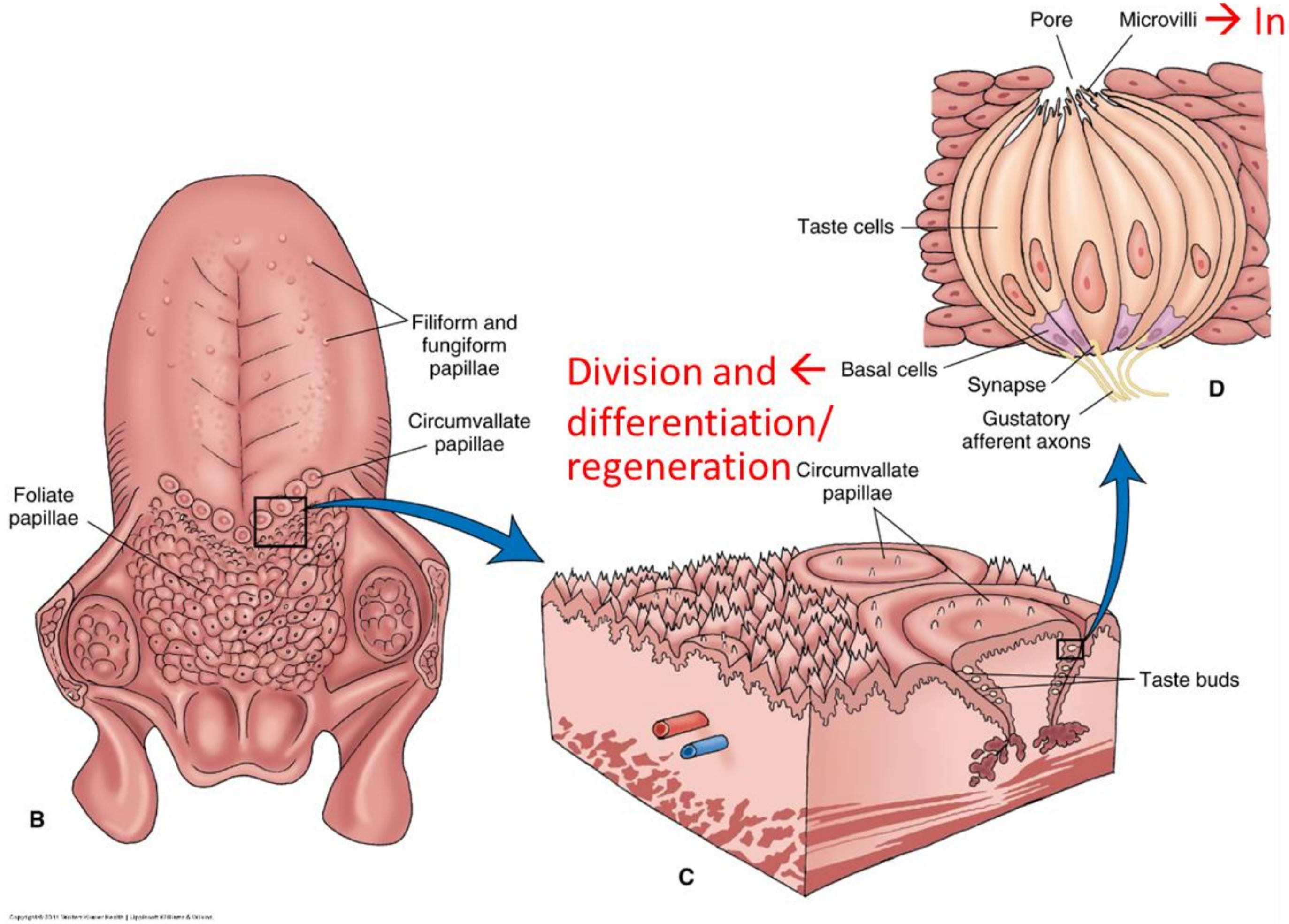
Basic Taste Qualities

- Salty (sodium chloride)
- Sour (acid)
- Sweet (sugar)
- Bitter (wide range of molecules, many toxic)

Fifth Taste Quality

- Umami (monosodium glutamate)

Components of the Gustatory System



→ Increase surface area (Transduction site)

Division and differentiation/ regeneration ←

→ CN VII (Anterior tongue)
→ CN IX (Posterior tongue)
→ CN X (Epiglottis)

Image from: Siegel, A and Sapru, HN, Essential Neuroscience, 4th edition © Wolters Kluwer Health (2019), ***Figure 17.4***

Gustatory Signal Transduction

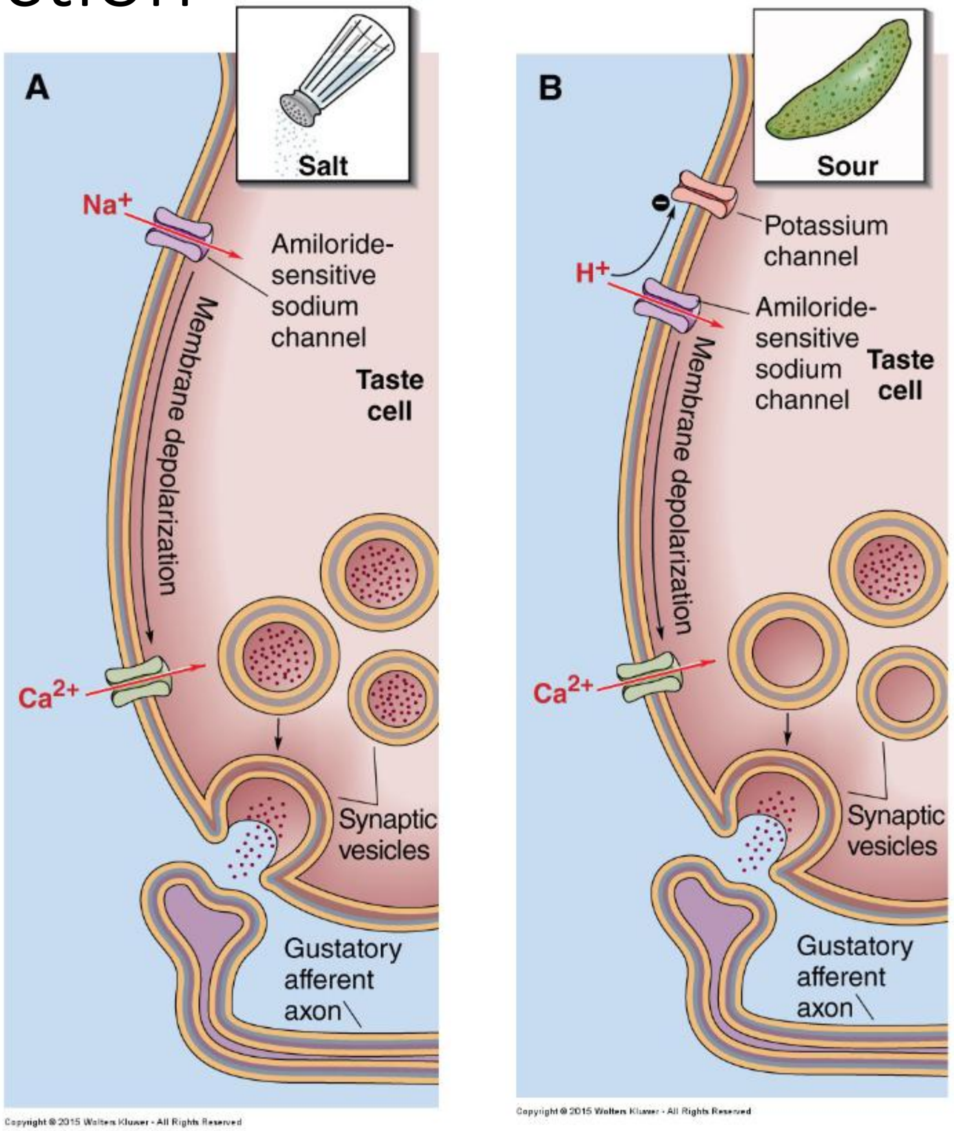
- Tastants may:
  - Directly pass through ion channels (salt and sour)
  - Bind to and block ion channels (sour)
  - Bind to G-protein-coupled receptors that lead to ion channel opening (bitter, sweet, umami; G-protein e.g. gustducin)

Gustatory Signal Transduction

Ionotropic signal transduction

- Salty (Na^+ ions)
- Sour (H^+ ions)

Taste ion → Current through ion channel → Depolarization → Transmitter release



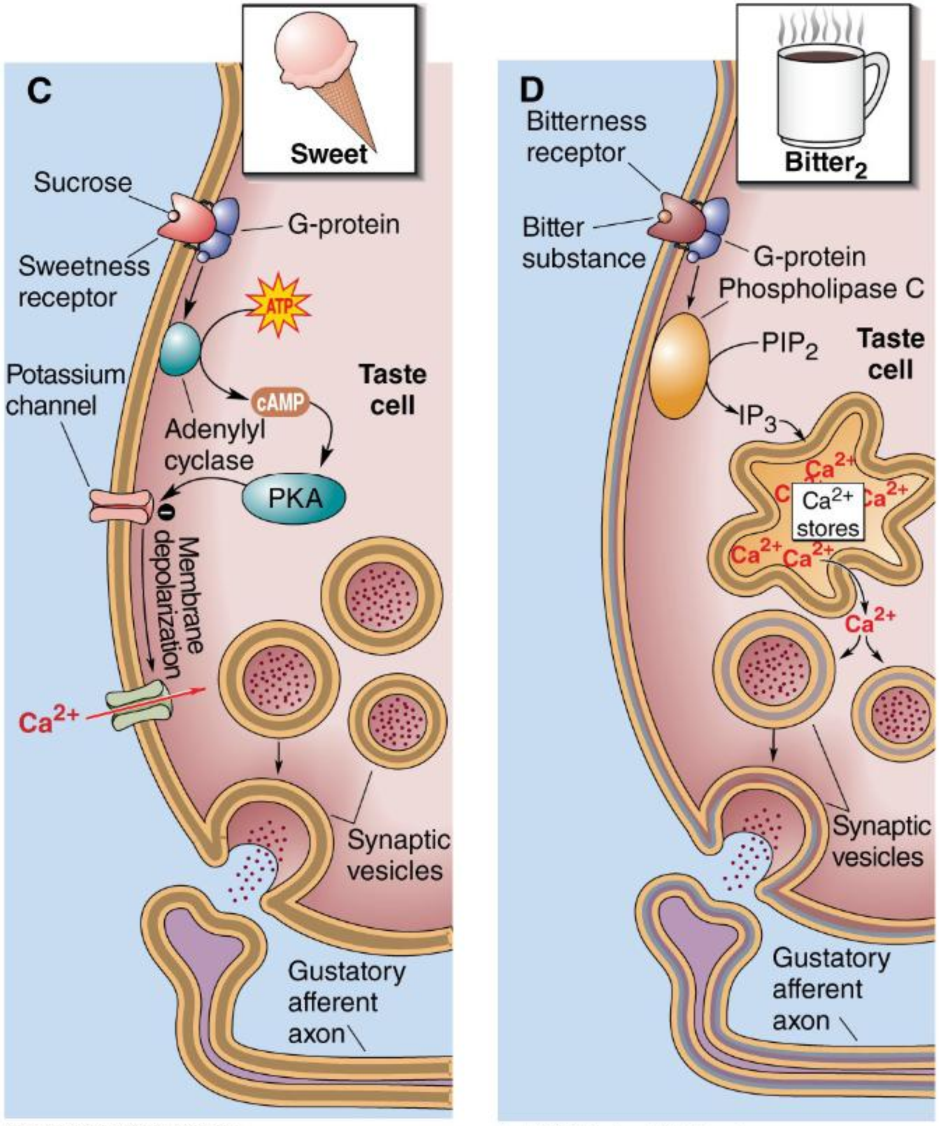
Images from: Siegel, A and Sapru, HN, Essential Neuroscience, 4th edition
© Wolters Kluwer Health (2019), ***Figure 17.5***

Gustatory Signal Transduction

Metabotropic signal transduction

- Sweet (second messenger cAMP)
- Bitter (second messenger IP_3)

Taste molecule → Receptor binding → G-protein → Enzyme → Second messenger → Current through ion channel → Depolarization → Transmitter release



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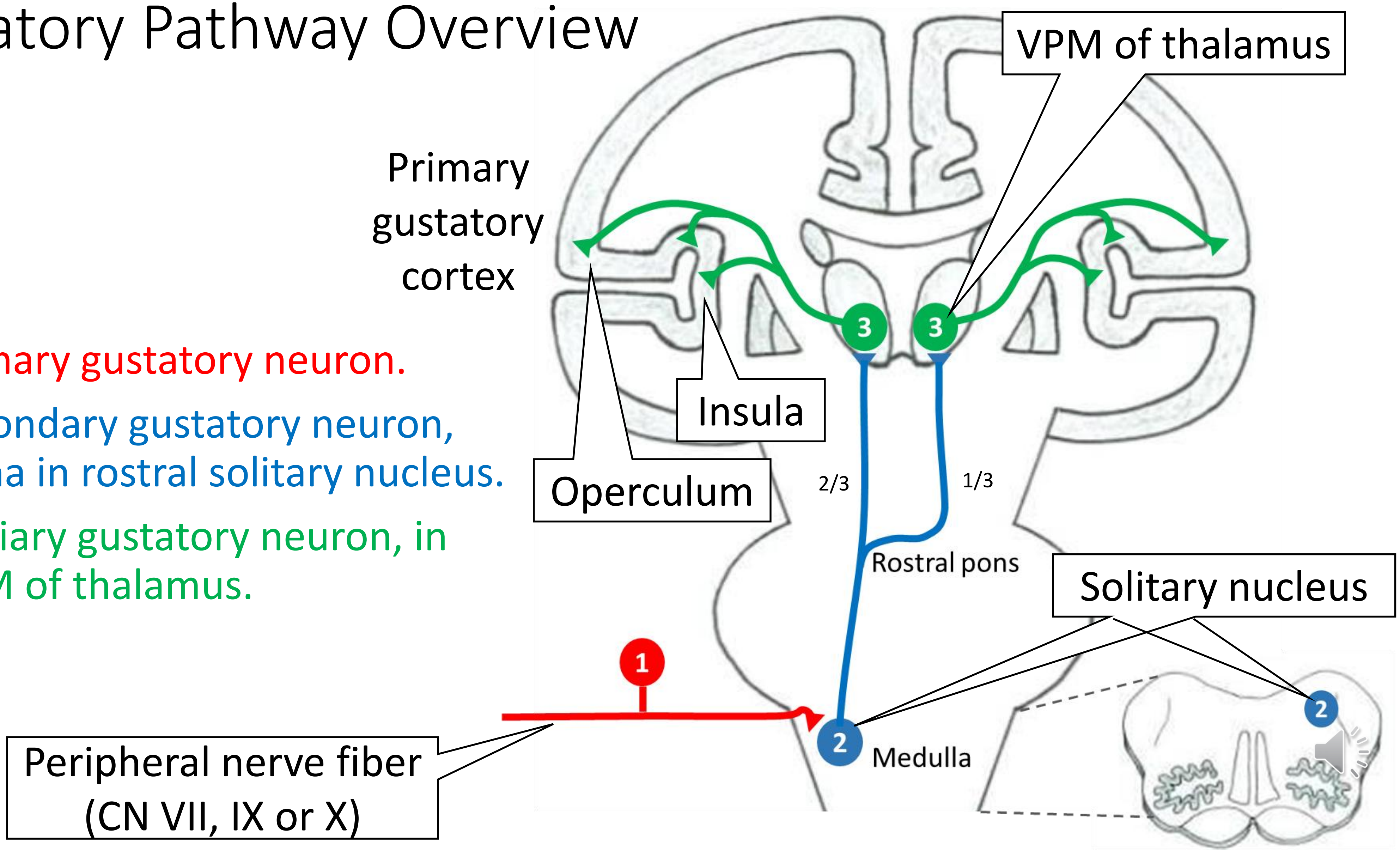
Gustatory Pathway Overview

1. Primary gustatory neuron.
2. Secondary gustatory neuron, soma in rostral solitary nucleus.
3. Tertiary gustatory neuron, in VPM of thalamus.



The End of Chemical Senses DLA: Gustation

If you have any questions, please send an email to:

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